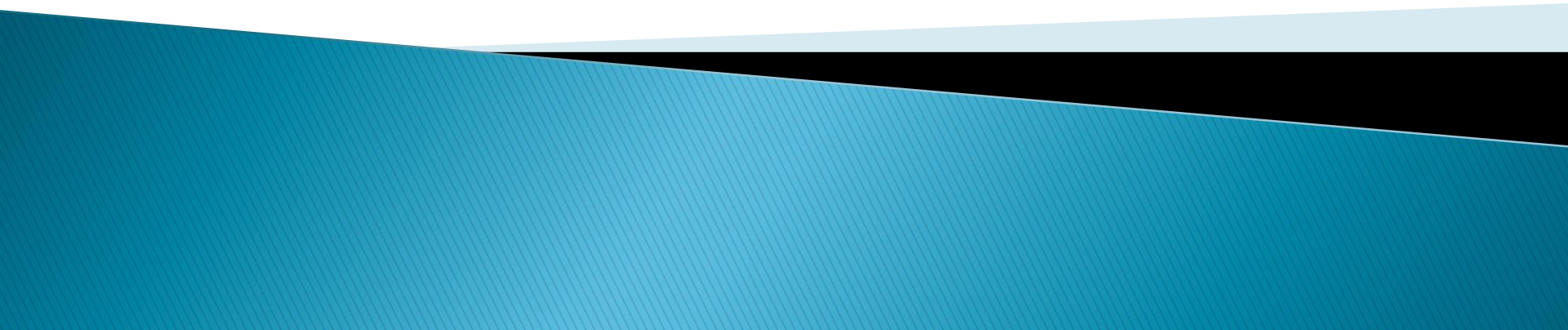


Chromatographic Methods of Separation CHEM 313

Dr. Reem M. Alghanmi

2017 1st term



Instructor Information

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Course Objectives

•The following objectives should be achieved by the students

•Solvent Extraction

- 1.The Distribution Coefficient
- 2.The Distribution Ratio
- 3.The Percent Extracted.
- 4.Solvent Extraction of Metals.
- 5.Analytical Separations.
- 6.Multiple Batch Extractions.
- 7.Countercurrent Distribution.
- 8.Solid-phase Extraction.
- 9.Solvent Extraction by Flow Injection Analysis.

•Chromatographic Methods

- 2.1 Principles of Chromatography.
- 2.2 Classifications of Chromatographic Techniques.
- 2.3 Techniques of Column Chromatography.
- 2.4 Column Efficiency in Chromatography.
- 2.5 Size Exclusion Chromatography.
- 2.6 Ion Exchange Chromatography.
- 2.7 Gas Chromatography.
- 2.9 Gas Chromatography-Mass Spectrometry.
- 2.10 High Performance Liquid Chromatography.
- 2.12 Paper Chromatography
- 2.14 Electrophoreses

Learning Resources

➤ QUANTITATIVE CHEMICAL ANALYSIS

D. Harris 8th ed.

➤ ANALYTICAL CHEMISTRY. Gary D. Christian. Fifth Ed. JOHN WILEY & SONS, INC.

➤ Note of Lab.

Assessment summary

Assessment Task	Due Date	Weighting
Exam 1	The sixth week	15%
Exam 2	The twelfth week	15%
Quizzes	The fourth, tenth and fifteenth weeks	10%
Lab	All weeks	20%
Final Exam	Examination period	40%
Summation		100%

✓CONTACT ME



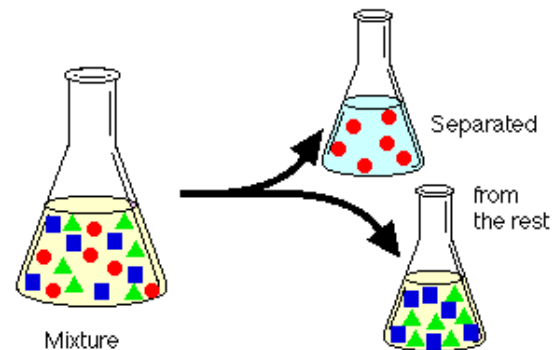
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✓منتدى المقرر في البلاك بورد

Solvent Extraction

- **Extraction** is the transfer of a solute from one phase to another.
- Solvent extraction involves the distribution of solute between two immiscible liquid phase.
- **The reasons** to use an extraction in analytical chemistry are to isolate or concentrate the desired analyte or separate it from species that would interfere in the analysis.
- This technique is extremely useful for very rapid and clean separations of both organic and inorganic substances.
- The most common case is the extraction of an aqueous solution with an organic solvent.



Diethyl ether, toluene, and hexane are common solvents that are *immiscible* with and less dense than water. They form a separate phase that floats on top of the aqueous phase. Chloroform, dichloromethane, and carbon tetrachloride are common solvents that are denser than water.[†] In the two-phase mixture, one phase is predominantly water and the other phase is predominantly organic.

[†]Whenever a choice exists between CHCl_3 and CCl_4 , the less toxic CHCl_3 should be chosen. Hexane and toluene are greatly preferred over benzene, which is a carcinogen.

Two liquids are **miscible** if they form a single phase when they are mixed in any ratio.

Immiscible liquids remain in separate phases.

Organic solvents with low polarity are generally immiscible with water, which is highly polar.

16.1. THE DISTRIBUTION COEFFICIENT

- A solute S will distribute itself between two phases after shaking and allowing the phase to separate and, within limits, **the ratio of the amounts of the solute in the two phases will be a constant**:

$$K_D = \frac{[S]_1}{[S]_2}$$

where:

- K_D is the distribution coefficient
- Solvent 1 is an organic solvent
- Solvent 2 is a water
- If the K_D large, the solute will tend toward quantitative distribution in solvent 1.

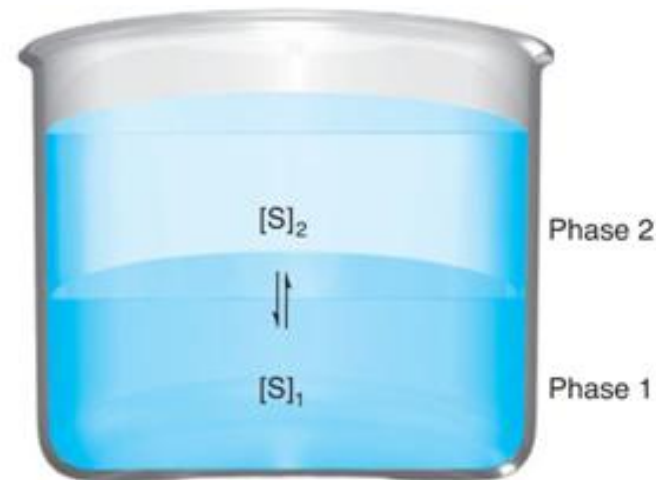
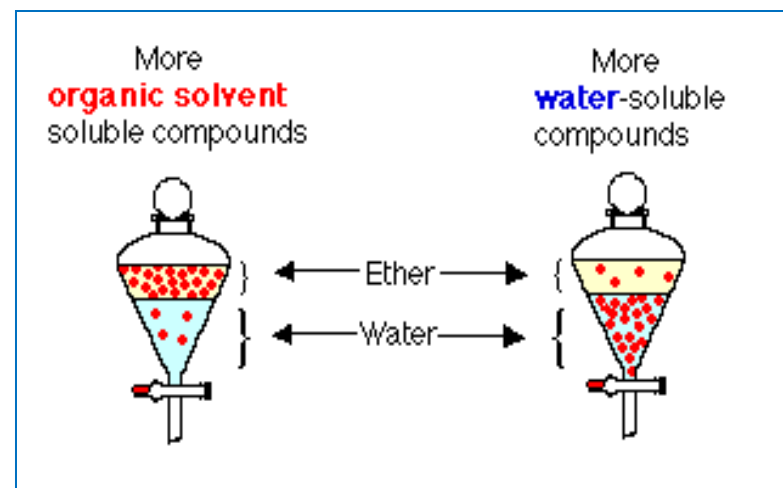


FIGURE 22-1 Partitioning of a solute between two liquid phases.



- The apparatus used for solvent extraction is **the separator funnel**.
- Often a solute is extracted from an aqueous solution into an immiscible organic solvent.
- After the mixture is shaken for about a minute, the phases are allowed to separate and the bottom layer (the denser solvent) is drawn off in a completion of the separation.

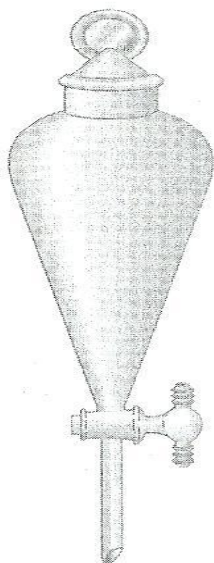
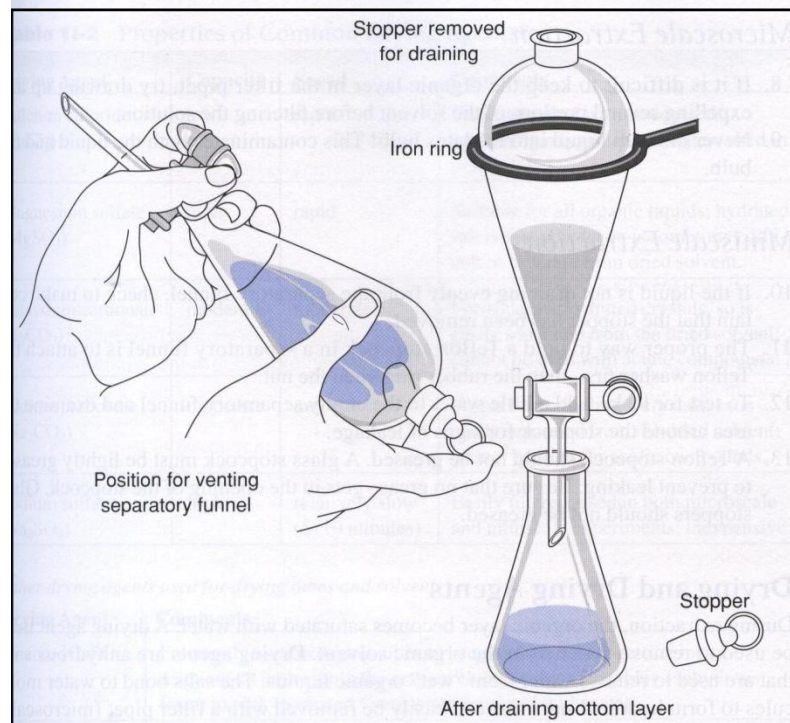


FIGURE 16.1 A separatory funnel.



- Many substances are partially ionized in the aqueous layers for example weak acids.
- This introduces a pH effect on the extraction.
- **For Ex. The extraction of benzoic acid from an aqueous solution.**
- Benzoic acid (BHz) is a weak acid in water with a particular ionization constant K_a .



- The distribution coefficient is

$$K_D = \frac{[\text{HBz}]_e}{[\text{HBz}]_a}$$

- Where **e** represents the ether solvent and **a** represents the aqueous solvent.
- Part of the benzoic acid in the aqueous layer will exist as Bz^- , depending on the magnitude of K_a and on the pH of the aqueous layer; hence, quantitative separation may not be achieved.

16.2. THE DISTRIBUTION RATIO

- The distribution ratio is **the ratio of the concentration of all the species of the solute in each phase.**

$$D = \frac{[S]_o}{[S]_a}$$

- In benzoic acid example it is given by

$$D = \frac{[HBz]_e}{[HBz]_a + [Bz^-]_a}$$

The relationship between D and K_D from the equilibria

The acidity constant K_a for the ionization of the acid in the aqueous phase is given by

$$K_a = \frac{[H^+]_a [Bz^-]_a}{[HBz]_a}$$

$$[Bz^-]_a = \frac{K_a [HBz]_a}{[H^+]_a}$$

$$K_D = \frac{[HBz]_e}{[HBz]_a}$$

$$[HBz]_e = K_D [HBz]_a$$

$$D = \frac{K_D [HBz]_a}{[HBz]_a + K_a [HBz]_a / [H^+]_a}$$

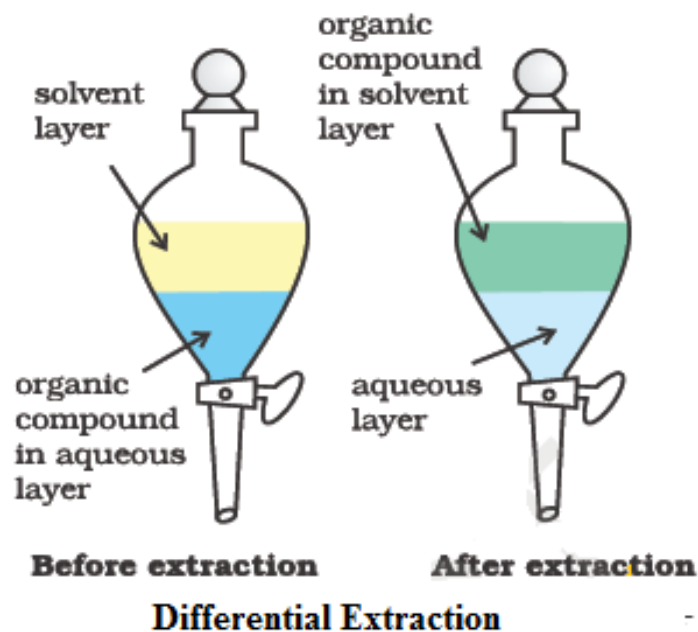
$$D = \frac{K_D}{1 + K_a / [H^+]_a}$$

$$D = \frac{K_D}{1 + K_a / [H^+]_a}$$

From the Eq. it is clear that:

- If $[H^+]_a \gg K_a$, D is nearly equal to K_D
- If K_D is large, the benzoic acid will be extracted into ether layer and D is maximum under these conditions.
- If, $[H^+]_a \ll K_a$, then D reduces to $K_D[H^+]_a/K_a$, which will be small, and the benzoic acid will remain in the aqueous layer.
- That is, **in alkaline** solution, the benzoic acid is ionized and cannot be extracted, while **in acid** solution, it is largely undissociated.
- These conclusions are what we would intuitively expect from inspection of the chemical equilibria.
- It is clear that the extraction efficiency will be **independent** of the original concentration of the solute.
- This is one of the attractive features of solvent extraction; it is applicable to tracer levels and to macro levels alike, a condition that applies only so long as the solubility of the solute in one of the phases is not exceeded and there are no side reactions such as dimerization of extracted solute.

- If **hydrogen ion concentration changes**, the extraction efficiency (D) will change.
- **In this example**, the hydrogen ion concentration will increase with increasing benzoic acid concentration, unless an acid-base buffer is added to maintain the hydrogen ion concentration constant.



16.3. THE PERCENT EXTRACTED

- The distribution ratio D is a constant independent of the volume ratio.
- The fraction of the solute extracted will depend on the volume ratio of two solvents.
- If a larger volume of organic solvent is used, more solute must dissolve in this layer to keep the concentration ratio constant and to satisfy the distribution ratio.
- $\text{fraction of solute extracted} = \frac{(\text{mmol of } S)_{\text{organic layer}}}{(\text{mmol of } S)_{\text{total}}}$
- The millimoles are given by the molarity times the milliliters.

$$\text{mmol} = \text{Molarity} \times \text{mL}$$

- The percent extracted is given by

$$\%E = \frac{[S]_o V_o}{[S]_o V_o + [S]_a V_a} \times 100\%$$

where V_o and V_a are the volumes of the organic and aqueous phases, respectively.

- It can be shown from this eq. that the percent extracted is related to the distribution ratio by

$$\%E = \frac{100D}{D + (V_a / V_o)}$$

- If $V_a = V_o$ then

$$\%E = \frac{100D}{D + 1}$$

- In the case of equal volumes, the solute can be considered quantitatively retained if D is less than 0.001. It is essentially quantitatively extracted if D is greater than 1000. The percent extracted changes only from 99.5% to 99.9% when D is increased from 200 to 1000.

Eq.

$$\%E = \frac{100D}{D + (V_a / V_o)}$$

- Shows that the fraction extracted can be increased by decreasing the ratio of V_a / V_o , for example, by increasing the organic phase volume.
- A more efficient way of increasing the amount extracted using the same volume of organic solvent is to perform successive extractions with smaller individual volumes of organic solvent.
- For example, with a D of 10 and $V_a / V_o = 1$, the percent extracted is about 91%. Decreasing V_a / V_o to 0.5 (doubling V_o) would result in an increase of $\%E$ to 95%. But performing two successive extractions with $V_a / V_o = 1$ would give an overall extraction of 99%.

EXAMPLE 16.1.

Twenty milliliters of an aqueous solution of 0.1 M butyric acid is shaken with 10 mL ether. After the layers are separated, it is determined by titration that 0.5 mmol butyric acid remains in the aqueous layer.

What is the distribution ratio, and what is the percent extracted?



- ▶ mmole extracted of butyric acid = $2 - 0.5 = 1.5$ mmol
- ▶ $[\text{Butyric acid}]_o = 1.5/10 = 0.15$ M
- ▶ $[\text{Butyric acid}]_a = 0.5/20 = 0.025$ M

$$D = \frac{[s]_o}{[s]_a} \quad \longrightarrow \quad D = \frac{0.15}{0.025} = 6$$

$$\%E = \frac{100D}{D + (V_a/V_o)} \quad \longrightarrow \quad \%E = \frac{100 \times 6.0}{6.0 + (20/10)} = 75\%$$

16.4. SOLVENT EXTRACTION OF METALS

- Solvent extraction has one of its most important applications in the separation of metal cations.
- This separation can be accomplished in several ways.
- The uncharged organic molecules tend to dissolve in the organic layer while the charge anion from the ionized molecules remains in the polar aqueous layer. **Like dissolves like.**
- Metal ions do not tend to dissolve appreciably in the organic layer.
- For them to become soluble, their charge must be neutralized and something must be added to make them **organic like.**

To extract a metal ion into an organic solvent, its charge must be neutralized, and it must be associated with an organic agent.

M^{n+} is in the aqueous phase and ML_n is in the organic phase.

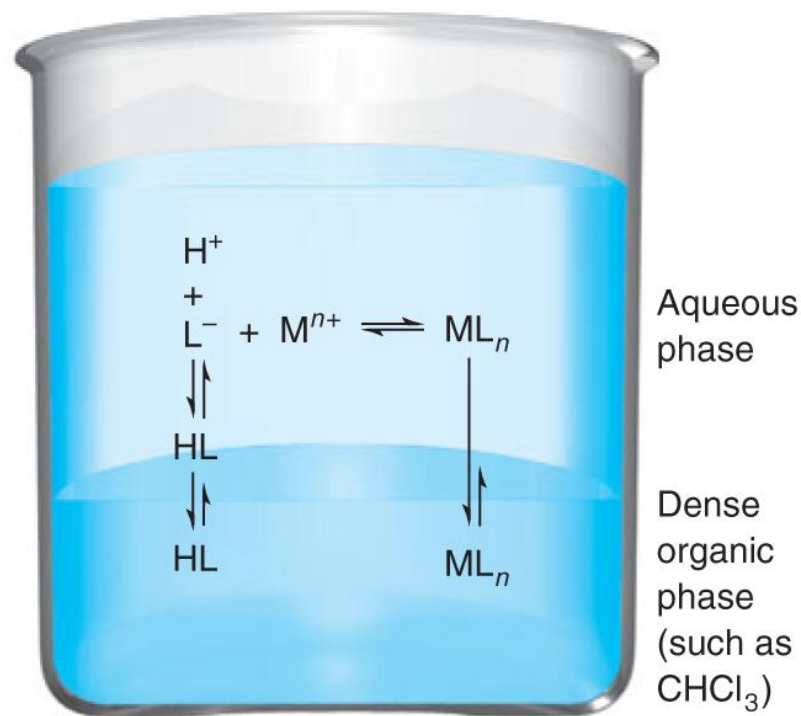


FIGURE 22-3 Extraction of a metal ion with a chelator. The predominant form of metal in the aqueous phase is M^{n+} , and the predominant form in the organic phase is ML_n .

The two principle solvent extraction systems for metal ions

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graph TD; A["The two principle solvent extraction systems for metal ions"] --> B["Ion-Association Complexes"]; A --> C["Metal Chelates"];
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**Ion-Association
Complexes**

Metal Chelates

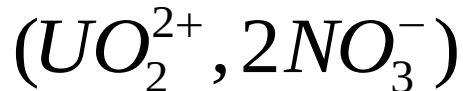
1- Ion-Association Complexes

- The metal ion is incorporated into a bulky molecule and then associates with ion of the opposite charge to form an ion pair, or the metal ion associates with another ion great size (organic-like).
- **Example: Iron(III)** can be quantitatively extracted from HCl acid medium into diethyl ether
- **The mechanism:**

The chloro complex of iron is coordinated with oxygen atom of the solvent (the solvent displaces the coordinated water), and this ion associates with a solvent molecule that is coordinated with proton:

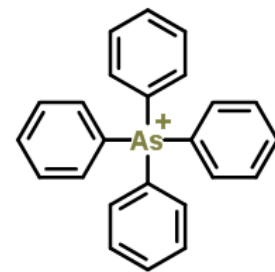


- **The uranyl ion UO_2^{2+}** is extracted from aqueous nitrate solution into isobutanol by associating with two nitrate ions

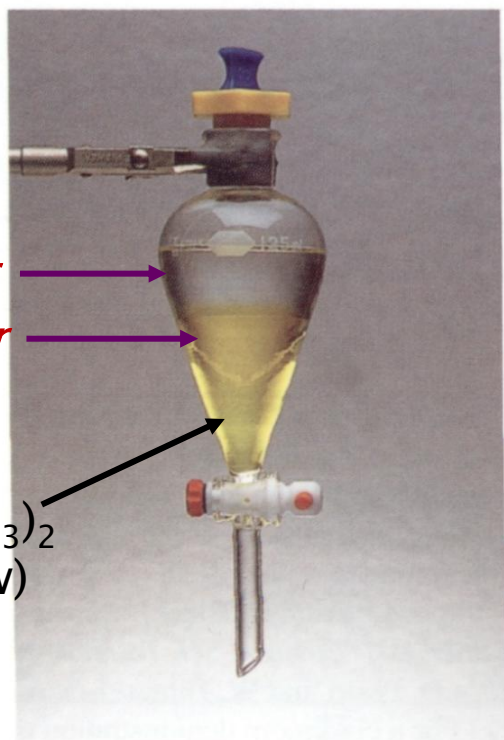


with the uranium probably being solvated by the solvent to make it solvent-like.

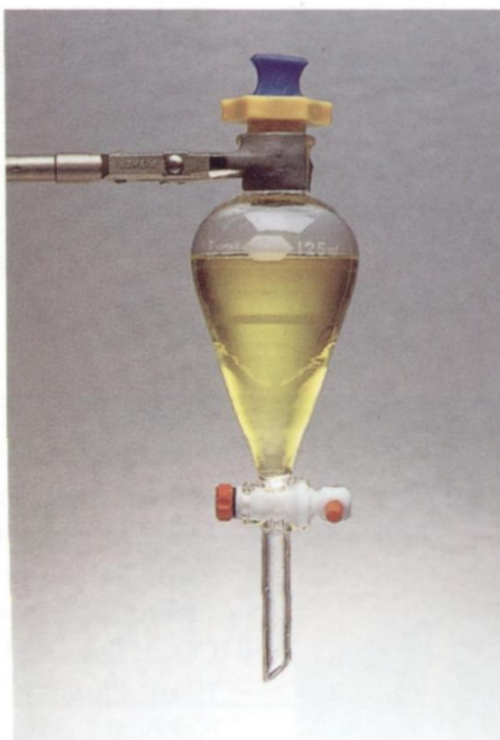
- **Permanganate** forms an ion pair with tetraphenylarsonium ion



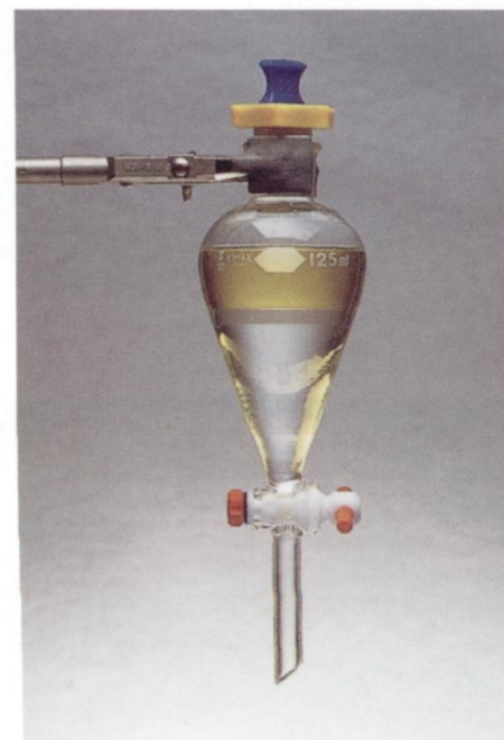
which makes it organic-like, and it is extracted into methylene chloride.



At the beginning



After mixing,
 $\text{UO}_2(\text{NO}_3)_2$
is distributed in
both layers



After 8 extractions,
 $\text{UO}_2(\text{NO}_3)_2$ has been
removed from water

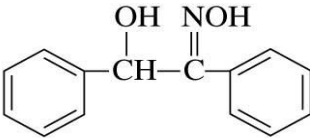
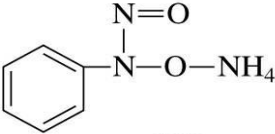
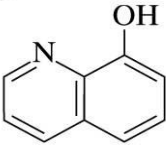
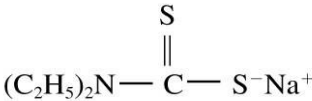
2- Metal Chelates

- The most widely used method of extracting metal ions is formation of a chelate molecule with an organic chelating agent.
- A chelating agent contains two or more complexing groups.
- Many of these reagents form colored chelates with metal ions and form the basis of spectrophotometric methods for determining the metals.
- The chelates are often insoluble in water and will precipitate.
- They are usually soluble in organic solvents such as chloroform, carbon tetrachloride or methylene chloride.
- Many of the organic precipitating agents are used as extracting agents.
- **Example:**
Diphenylthiocarbzone (dithizone), which forms a chelate with lead ion.

Organic precipitating agents are chelating agents.
They form insoluble metal chelates.

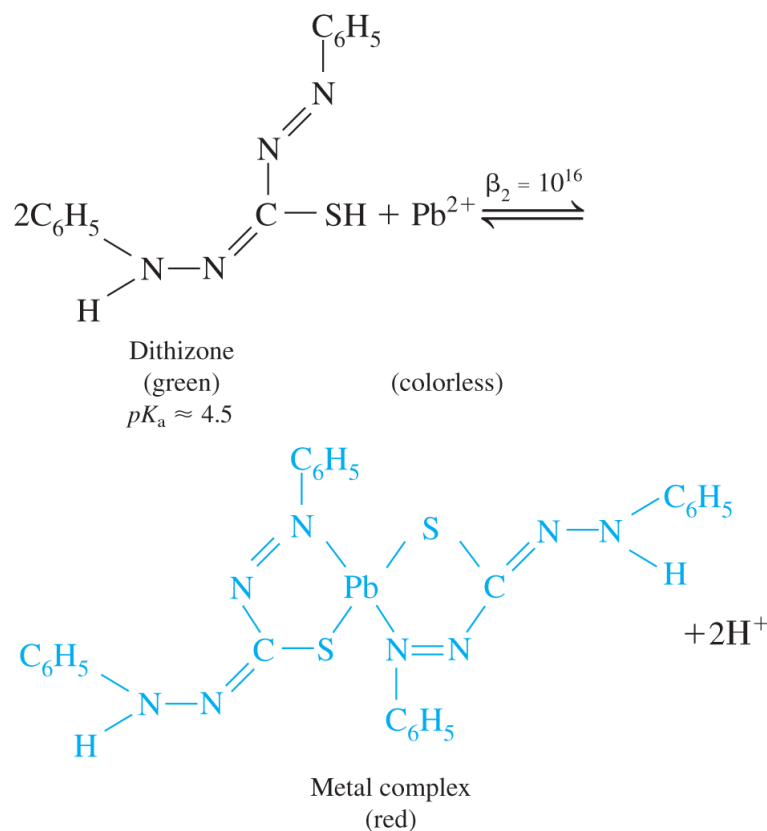
Table 10.2

Some Organic Precipitating Agents

Reagent	Structure	Metals Precipitated
Dimethylglyoxime	$\text{CH}_3 - \text{C} = \text{NOH}$ $\text{CH}_3 - \text{C} = \text{NOH}$	Ni(II) in NH_3 or buffered HOAc; Pd(II) in HCl $(\text{M}^{2+} + 2\text{HR} \rightarrow \text{MR}_2 + 2\text{H}^+)$
α -Benzoinoxime (cupron)		Cu(II) in NH_3 and tartrate; Mo(VI) and W(VI) in H^+ $(\text{M}^{2+} + \text{H}_2\text{R} \rightarrow \text{MR} + 2\text{H}^+; \text{M}^{2+} = \text{Cu}^{2+}, \text{MoO}_2^{2+}, \text{WO}_2^{2+})$ Metal oxide weighed
Ammonium nitrosophenylhydroxylamine (cupferron)		Fe(III), V(V), Ti(IV), Zr(IV), Sn(IV), U(IV) $(\text{M}^{n+} + n\text{NH}_4\text{R} \rightarrow \text{MR}_n + n\text{NH}_4^+)$ Metal oxide weighed
8-Hydroxyquinoline (oxine)		Many metals. Useful for Al(III) and Mg(II) $(\text{M}^{n+} + n\text{HR} \rightarrow \text{MR}_n + n\text{H}^+)$
Sodium diethyldithiocarbamate		Many metals from acid solution $(\text{M}^{n+} + n\text{NaR} \rightarrow \text{MR}_n + n\text{Na}^+)$
Sodium tetraphenylboron	$\text{NaB}(\text{C}_6\text{H}_5)_4$	$\text{K}^+, \text{Rb}^+, \text{Cs}^+, \text{Tl}^+, \text{Ag}^+, \text{Hg(I)}, \text{Cu(I)}, \text{NH}_4^+, \text{RNH}_3^+, \text{R}_2\text{NH}_2^+, \text{R}_3\text{NH}^+, \text{R}_4\text{N}^+$. Acidic solution $(\text{M}^+ + \text{NaR} \rightarrow \text{MR} + \text{Na}^+)$
Tetraphenylarsonium chloride	$(\text{C}_6\text{H}_5)_4\text{AsCl}$	$\text{Cr}_2\text{O}_7^{2-}, \text{MnO}_4^-, \text{ReO}_4^-, \text{MoO}_4^{2-}, \text{WO}_4^{2-}, \text{ClO}_4^-, \text{I}_3^-$. Acidic solution $(\text{A}^{n-} + n\text{RCl} \rightarrow \text{R}_n\text{A} + n\text{Cl}^-)$

DEMONSTRATION 22-1 Extraction with Dithizone

Dithizone (diphenylthiocarbazone) is a green compound that is soluble in nonpolar organic solvents and insoluble in water below pH 7.³ In alkaline aqueous solution, it forms a soluble yellow ion. It forms red, hydrophobic complexes with di- and trivalent metals. Dithizone is used for analytical extractions and for colorimetric determinations of metal ions.



You can demonstrate the equilibrium between green ligand and red complex by using three large test tubes sealed with tightly fitting rubber stoppers. Place some hexane plus a few milliliters of dithizone solution (prepared by dissolving 1 mg of dithizone in 100 mL of CHCl_3) in each test tube. Add distilled water to tube A, tap water to tube B, and 2 mM $\text{Pb}(\text{NO}_3)_2$ to tube C. After shaking and settling, tubes B and C contain a red upper phase, whereas A remains green.

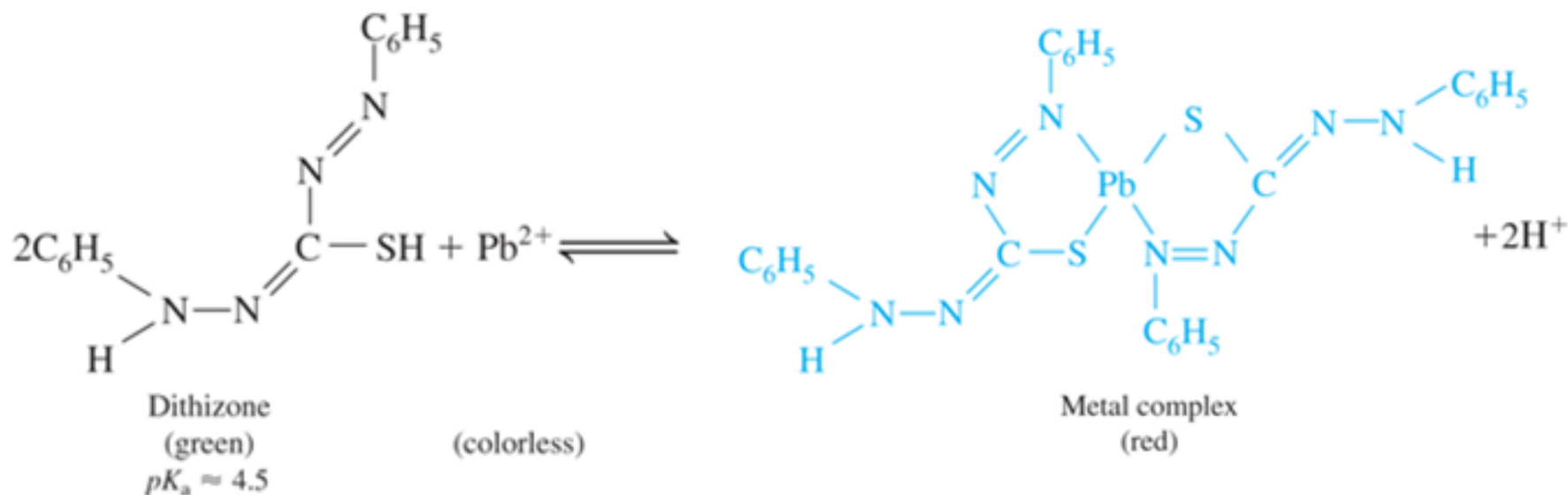
Proton equilibrium in the dithizone reaction is shown by adding a few drops of 1 M HCl to tube C. After shaking, the dithizone turns green again. Competition with a stronger ligand is shown by adding a few drops of 0.05 M EDTA solution to tube B. Again, shaking causes a reversion to the green color.

Practicing “Green” Chemistry

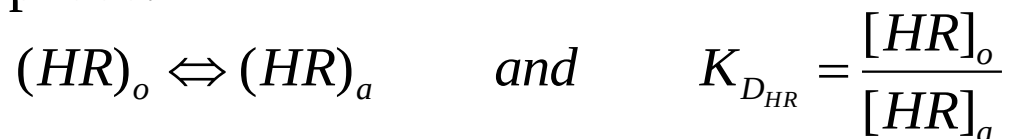
Chemical procedures that produce less waste or less hazardous waste are said to be “green” because they reduce harmful environmental effects. In chemical analyses with dithizone, you can substitute aqueous micelles (Box 25-1) for the organic phase (which has traditionally been chloroform, CHCl_3) to eliminate chlorinated solvent and the tedious extraction.⁴ For example, a solution containing 5.0 wt% of the micelle-forming surfactant Triton X-100 dissolves 8.3×10^{-5} M dithizone at 25°C and $\text{pH} < 7$. The concentration of dithizone inside the micelles, which constitute a small fraction of the volume of solution, is much greater than 8.3×10^{-5} M. Aqueous micellar solutions of dithizone can be used for the spectrophotometric analysis of metals such as Zn(II), Cd(II), Hg(II), Cu(II), and Pb(II) with results comparable to those obtained with an organic solvent.

The Extraction Process

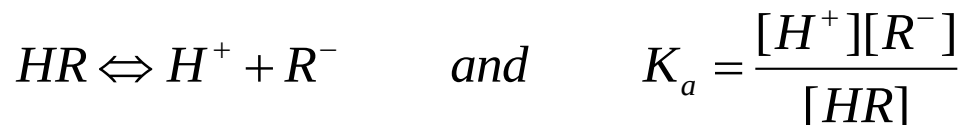
- Most chelating agents are weak acids that ionize in water; the ionizable proton is displaced by the metal ion when the chelate is formed
- The charge on the organic compound neutralizes the charge on the metal ion.
- **Example:**
- Diphenylthiocarbazone (dithizone), forms a chelate with lead ion.



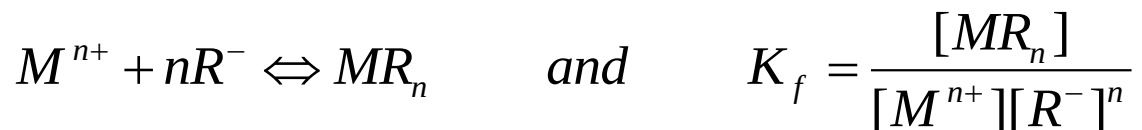
- The usual practice is to add the chelating agent to the organic phase.
- The extraction process can be thought to consist of four equilibrium steps, each with an equilibrium constant.
- **First:** the chelating agent HR distributes between the aqueous and the organic phase.



- **Second:** the reagent in the aqueous phase ionizes



- **Third:** the metal ion chelates with the reagent anion to form an uncharged molecule



- **Finally:** the chelate distributes between the organic and aqueous phases

$$(MR_n)_a \rightleftharpoons (MR_n)_o \text{ and } K_{D_{MRn}} = \frac{[MR_n]_o}{[MR_n]_a}$$

- $K_{D_{HR}}$ and $K_{D_{MRn}}$ are the distribution coefficients of the reagent and the chelate, respectively
- K_a is the ionization constant of the reagent
- K_f is the formation constant of the chelate.

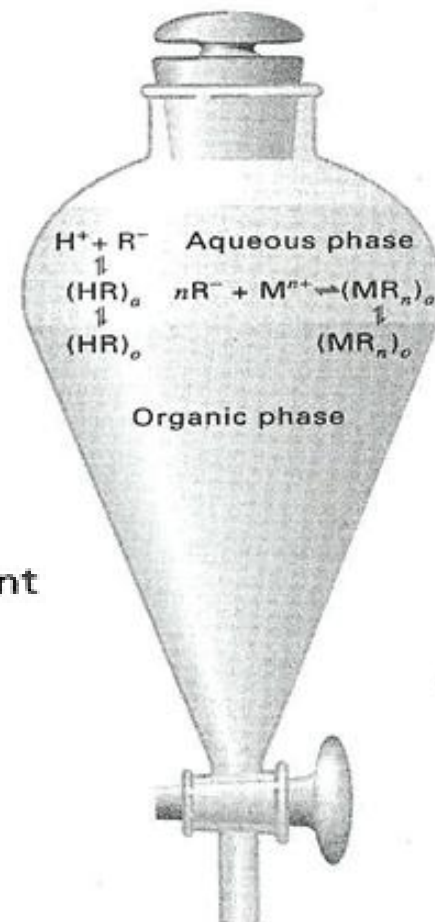


FIGURE 16.2 Equilibria involved in the solvent extraction of metal chelates.

- Assuming that the related portion of the metal distributes largely into the organic phase, the metal ion does not hydrolyze in the aqueous phase, and the chelate is essentially undissociated in the nonpolar organic solvent, the distribution ratio is given by

$$D = \frac{[MR_n]_o}{[M^{n+}]_a}$$

- By substituting the previous equilibrium concentrations into the above Eq., so that all four equilibrium constants are included, the following Eq. can be derived:

$$D = \frac{K_{D_{MRn}} K_f K_a^n}{K_{D_{HR}}^n} \cdot \frac{[HR]_o^n}{[H^+]_a^n} = K \cdot \frac{[HR]_o^n}{[H^+]_a^n}$$

$$D = \frac{K_{D_{MRn}} K_f K_a^n}{K_{D_{HR}}^n} \cdot \frac{[HR]_o^n}{[H^+]_a^n} = K \cdot \frac{[HR]_o^n}{[H^+]_a^n}$$

From the Eq. it is clear that:

- The distribution ratio is independent of the concentration of the solute (metal ion).
- The extraction efficiency can be effected only by **changing the reagent concentration** or by **changing the pH**.
- If the reagent concentration **increases**, the extraction efficiency will **increases** by the same amount as an increase in the pH of one unit.
- These effects are greater as n becomes larger.
- The more stable the chelate (the larger the K_f), the greater the extraction efficiency.

The Separation Efficiency of Metal Chelates

The separation efficiency of two metals at a given pH and reagent concentration can be predicted from

$$D = \frac{K_{D_{MRn}} K_f K_a^n}{K_{D_{HR}}^n} \cdot \frac{[HR]_o^n}{[H^+]_a^n} = K \cdot \frac{[HR]_o^n}{[H^+]_a^n}$$

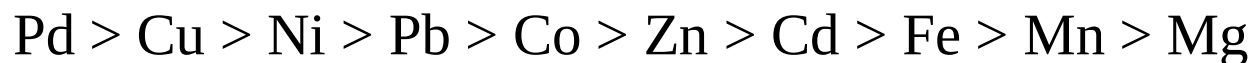
The separation factor β is equal to the ratio of the distribution ratios of the two metal chelates formed with a given reagent.

Since only K_f and $K_{D_{MRn}}$ will be a function of the metal, then

$$\beta = \frac{D_1}{D_2} = \frac{K_{f(1)} K_{D_{MRn}(1)}}{K_{f(2)} K_{D_{MRn}(2)}}$$

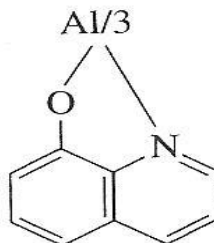
So, the separation efficiency depends on the relative formation constants and on the relative solubilities of the chelates.

- The order of stability of complexes of limited number of divalent metals has been shown, other things being equal, to be fairly independent of the nature of the chelating agents, with the following stability sequence:

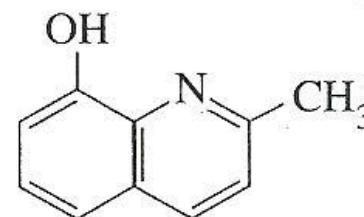


- The order of extraction may be altered from this by differences in solubility of the chelates.
- Also, **steric hindrance** (in which a functional group on the reagent molecule may hinder its reaction with a given metal ion) may affect the specificity of extraction.

- **For example:** 8-hydroxyquinoline (oxine) forms a chelate with many metals, including aluminum:



- The oxine derivative 2-methyl-8-hydroxyquinoline



will not form a complex with aluminum because the added methyl group does not allow room for three molecules to group around this small ion.

Therefore, other metals can be extracted in the presence of aluminum with this reagent.

Masking Agents

- They are added to increase the selectivity of extractions.
- These are competing complexing agents that form charged complexes that are more stable for certain metals.
- **Examples:**
- EDTA and cyanide ion are used as masking agents.
- Cu^{2+} forms a more stable complex with oxine than VO_2^{2+} , the vanadium may be extracted in the presence of Cu^{2+} by adding EDTA, which forms an even more stable complex with the copper (Cu-EDTA^{2-}).

Masking agents form charged complexes with interfering metals and prevent their extraction.

- The selectivity of an extraction can often be controlled by proper pH adjustment.

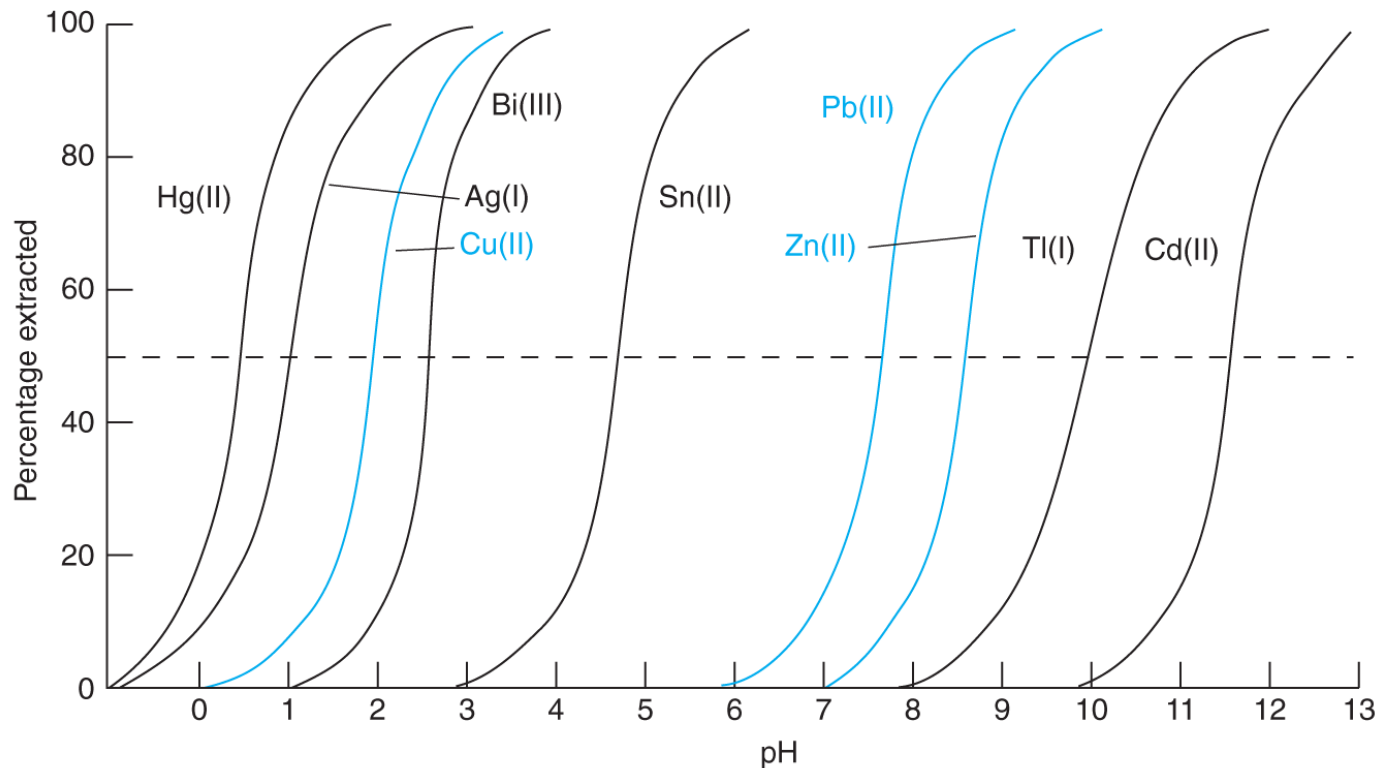


FIGURE 22-4 Extraction of metal ions by dithizone into CCl_4 . At pH 5, Cu^{2+} is completely extracted into CCl_4 , whereas Pb^{2+} and Zn^{2+} remain in the aqueous phase. [Adapted from G. H. Morrison and H. Freiser in C. L. Wilson and D. Wilson, eds., *Comprehensive Analytical Chemistry*, Vol. 1A (New York: Elsevier, 1959).]

pH control is an effective way of improving the selectivity of extraction.

16.5 ANALYTICAL SEPARATIONS

- One of the most important applications of solvent extraction is the spectrophotometric determination of metals in the visible region.
- Many organic reagents form colored chelates with metals, **but** most chelates are insoluble in water.
- They are soluble in organic solvents and can be extracted.
- Since the chelating agent is often itself colored, its absorption spectrum may overlap that of the metal chelate, and a blank correction must be made or the reagent washed from the organic phase.
- The selectivity of the determination can be adjusted by the factors discussed.
- Solvent extraction is widely employed in drug analysis.

➤ **Example:**

Separation and determination of lead in blood sample:

- 1- the organic matter is destroyed.
- 2- the dithizone chelate of lead is formed.
- 3- the chelate is extracted at pH 8-10 into methylene chloride.
- 4- it is determined spectrophotometrically.

➤ **Improve selectivity by use masking agent.**

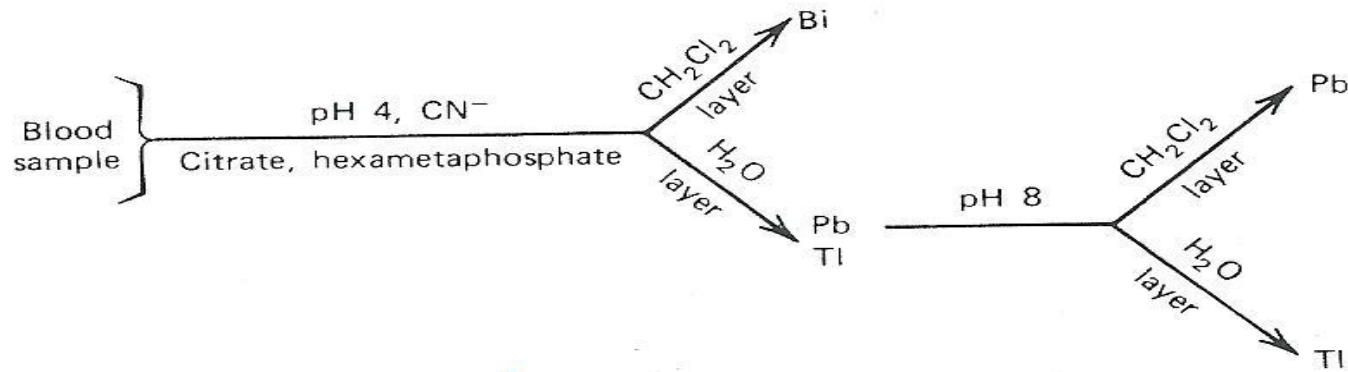


FIGURE 16.4 * Selective solvent extraction of lead from blood into methylene chloride using dithizone reagent.

Masking agents combined with pH control is even more effective for achieving selectivity.